

Computer Security & Cryptography Exam

Course Code: MATH319 | Total Marks: 20

Instructor: Dr. Sofiane Medjram **Semester:** Term 472 (2nd Semester, 1447 AH) **Chapters:** 1, 2, 3 (Up to Slide 51 - DES)

Question 1: True or False (4 items, 2 marks)

1. Stream ciphers are designed to encrypt a block of data at a time.

Answer: False

2. [Forgotten by student]

()

3. Symmetric encryption uses different keys for both encryption and decryption.

Answer: False

4. Modern technology has rendered the Data Encryption Standard (DES) insecure because its 56-bit key length is vulnerable to brute-force attacks.

Answer: True

Question 2: Multiple Choice Questions (2 items, 2 marks)

1. Which type of cryptanalytic attack is used when the attacker has access to both the ciphertext and its corresponding plaintext?

- a) Ciphertext-only attack
- b) Known-plaintext attack (Correct)**
- c) Chosen-plaintext attack
- d) Man-in-the-middle attack

2. In the DES (Data Encryption Standard) algorithm, what is the size of the input data block?

- a) 32 bits
- b) 56 bits
- c) 64 bits (Correct)**
- d) 128 bits

Question 3: Short Answer Questions (4 items, 8 marks)

1. What are Shannon's fundamental principles for designing a secure cipher, often implemented in S-P networks?

Answer: Substitution (to achieve confusion) and Permutation (to achieve diffusion).

2. Briefly distinguish between Block Ciphers and Stream Ciphers.

Answer: Block Ciphers encrypt data in fixed-size blocks (e.g., 64 bits), while Stream Ciphers encrypt data continuously.

3. What is cryptography?

Answer: It is the practice and study of techniques for securing communication and data against adversaries, ensuring confidentiality, integrity, and authenticity.

4. Why is the DES algorithm considered insecure in modern times?

Answer: Due to its short key length (56 bits), which can be broken by modern computers using brute-force attacks.

Question 4: Essay / Analysis Questions (3 items, 8 marks)

1. If a brute-force attack can successfully overcome an encryption key, what is a simple solution that can be applied to the key to enhance security?

Answer: Increase the key length (e.g., from 56-bit to 128 or 256 bits).

2. If a company grows from 5 employees to 1,000, why does using Symmetric Encryption become a bad idea for communication?

Answer: Key Management problem. The number of unique keys required $[n(n-1)/2]$ becomes impractical.

3. If we want to decrypt a ciphertext that was encrypted using the DES algorithm, do we need to use a different algorithm? Why or why not? (4 marks)

Answer: No. DES is a symmetric algorithm; encryption and decryption use the same logic, only the sub-key schedule is reversed during decryption.